

Samsung Galaxy S24

A minor upgrade with a sprinkle of AI magic

The Samsung Galaxy S24 undoubtedly shines in 2024, particularly for users upgrading from devices several generations old. For those less interested in AI and for those who possess a fairly recent Samsung flagship phone, the hardware remains largely unchanged. So, is it still worth your money? Let's delve into it.

BUILD AND DESIGN

The Samsung Galaxy S24 stands out as a premium and classy compact smartphone in an era where such form factors are becoming increasingly rare. The Galaxy S24 has the same dimensions as its predecessor but still manages to incorporate a larger display thanks to the bezels getting narrower this time around. The design refresh also features a flat matte aluminium frame that feels luxurious; and very iPhone-esque, might we add. There are slight contours in the shape though which makes it pretty comfortable to hold. Plus, the S24 is super lightweight!

The Amber Yellow unit we got for review has a very pastel, muted yellow tone. It is nowhere near as vibrant as the colour tone shown on Samsung's own website, but I would say the understated nature of these colours looks very classy. The matte rear resists fingerprints, but it is quite slippery, so use it with a nice and grippy case.

The ultrasonic fingerprint sensor is faster than I've ever experienced on a Samsung device before, although it may have trouble working with tempered glass screen protectors.

DISPLAY

Samsung has fitted a larger 6.2-inch display in the same footprint as the S23 this



time around. One notable upgrade in the display department is the improved brightness – using a LUX meter, we measured close to 2,200 nits of peak brightness in the Auto Brightness mode under bright sunlight, which is undoubtedly impressive.

The Galaxy S24's LTPO panel supports dynamic refresh rate. I noticed it going as low as 24 Hz and switching to 50, 60, 70, 80, 90, and 120 Hz based on the app being used at the time. The Always-On display brings the refresh rate down to 24 Hz. Sadly, I did not experience it going down to 1 Hz (through the Android refresh rate overlay).

Speaking of the Always On Display (AOD), Samsung has made a change this year. Like the iPhone 15 series, the AOD carries over the lock screen wallpaper and just dims it significantly when the screen is turned off. However, this results in higher battery consumption.

This is a world-class display with fantastic, vibrant colours (although there's still no 10-bit support), deep blacks, excellent viewing angles, and sharp details.

PERFORMANCE

Chipset fragmentation on the flagship Samsung smartphones is, unfortunately, back. In India, the Samsung Galaxy S24 and S24+ feature Samsung's own Exynos 2400 SoC while the Snapdragon 8 Gen 3 is reserved for the S24 Ultra. Samsung has maintained that both chipsets have equivalent performance chops, and I put that to the test as soon as I got my hands on the Exynos 2400-powered Galaxy S24. Now, let's get to the Exyn... ahem... elephant in the room. Does the Exynos 2400 chipset match the performance of the Snapdragon 8 Gen 3 for Galaxy?

In CPU benchmarks such as AnTuTu, there is a marginal difference. The S24 scores more than 1.6 million but flagships such as the OnePlus 12, iQOO 12, and the S24 Ultra do pull ahead. The S24 Ultra has an approximately 10 per cent higher score than the S24 in AnTuTu. In GeekBench, the only flagship the S24 could beat was the OnePlus 12. Other phones with the Snapdragon 8 Gen 3 pulled ahead, but strangely the OnePlus 12 didn't. The Galaxy S24 got a solid score of 17,736 in PCMark Work, meaning it will excel in performing tasks such as working on Excel sheets, editing photos, etc.

In GPU benchmarks, once again the S24 lagged behind the iQOO 12 and Galaxy S24 Ultra but it did outperform the OnePlus 12. So, as per benchmarks, there is a performance disparity between the Exynos 2400 and SD 8 Gen 3 but it is not as pronounced as it used to be the last time we had chipset fragmentation with the Galaxy series in India. There's only a 7-12 per cent uptick in performance on the Snapdragon 8 Gen 3 S24 Ultra.

However, where it falters is throttling. In the CPU Throttling Test, the Exynos 2400-powered Galaxy S24 throttled to 58 per cent of its peak performance, meanwhile, the S24 Ultra with the Snapdragon 8 Gen 3 only throttled to 73 per cent, which is a big difference. The S24 also runs 4-5 degrees (Celsius) hotter than the S24 Ultra, although this also boils down to the fact that the S24 Ultra has a more robust cooling system.

Now, the throttling really comes into play when you're gaming, video editing, or